Final Project Report

Computational Modelling of Social Systems

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1 Introduction

'Mindset', 'Catcalling' and 'framen' are only a few examples of English loanwords published in the latest edition of the German Duden (2024). Anglicisms and the influence of English words on the German language have been extensively studied, going back to Carstensen's (1965) work, and now, with the increased use of social media, the way we as a society communicate, share information, and interact with each other has changed and shows no sign of stopping.

Poplack and Sankoff (1984) have researched and argued that loanword adoption is not merely a linguistic phenomenon, but a social one which is driven by community and communication patterns. Looking at this from the German language perspective, previous work has focused more on corpus-based research and looking at how and what kind of English words are borrowed or adopted into the German language (Coats 2019). However, these approaches have mainly captured the outcomes of adoption rather than the social dynamics driving it.

More recently, sociolinguistic dynamics of diffusion on social media platforms have been examined (Würschinger 2021), giving light to the fact that network structure plays a role in determining whether new loanwords spread broadly or remain within a specific community. While this research was based solely on a single social media platform and not in the German context, this paper wants to answer the following research question: Does cross-platform social media usage accelerate the spread of English loanwords in German social media networks?

By implementing an agent-based model (ABM), we want to better understand how using multiple social media platforms facilitates the spread of new English loanwords from a computational approach. We draw from both Baronchelli et al.'s (2006) model of communication and Granovetter's (1973) threshold model of collective behavior to simulate adoption as a social contagion process where agent-level attributes such as varying levels of English proficiency and different generational groups are used to examine the spread and diffusion of anglicisms.

After running multiple rounds of simulations, it becomes evident, that starting off with several initial adopters distributed across different positions in the network leads to an accelerated overall adoption rate. The model also highlights, that the specific positioning of the initial adopters matters, since a cross-platform influencer drives a quicker and more synchronized adoption across all platforms compared to a randomly placed initial adopter. These findings suggest that social media platform membership can be an important variable in understanding language evolution in the digital space.

2 Methodology

To help us find the answer to our research question, we implemented an agent-based model using the Mesa framework for Python. In our model, each agent represents a German-speaking social media user with multiple different attributes that influence their adoption rate of a new English word. These attributes are the following: age group, English proficiency, main topic of interest, and used social media platforms. For the age group attribute, each agent belongs to exactly one of three categories: Gen Z, Millennial, or Boomer. The English proficiency has three values, ranging from low to high. Each agent also has a primary area of interest from the Tech, Fashion, Gaming, and Nature categories. Finally, all agents are part of one or multiple social media platforms, such as Reddit, TikTok, and Instagram. This gives us the ability to have agents that act as bridges between different platforms, so words used on only one platform have a chance to be adopted by other platforms.

Our simulated social media network is represented by an undirected network where each agent is added as a node and edges between them represent possible contact with the language used by another user. It is only possible for two agents to be connected if they share one or more social media platforms, because we assume that interaction between social media users happens mostly on the same platform. Agents who are active on multiple social media platforms are able to have connections to agents from multiple platforms and enable cross-platform spread of language. Each pair of agents with at least one platform in common has a fixed connection probability, which is set to 0.01. Our resulting network contains clusters based on social media platforms, with multi-platform users acting as links between those clusters.

The adoption of an English word in our model happens through social contagion. At the start of our simulation, we select a fixed number of random agents as our initial adopters. These agents simulate users who are already using the English word. During each simulation, all previous non-adopters evaluate if the word gets adopted in this round. The adoption probability is influenced by multiple different factors. We start with a base probability of 0.001, and if the agent has neighbors, the fraction of neighbors that have already adopted the word gets calculated, multiplied by 0.1, and added to the agent's adoption probability. The age group of the agent further changes the probability by adding 0.01 for Gen Z, 0.005 for Millennials, and nothing for Boomers. Furthermore, the agent's English proficiency adds another 0.001 for each level. The final adoption rate is capped at 1, which equates to 100% chance of adoption. If a randomly drawn decimal number between 0 and 1 is smaller than the adoption probability of the agent, the agent adopts the word and remains an adopter for the rest of the simulation.

Our main outcome after each simulated round is the global adoption rate, which describes the percentage of agents that have adopted the English word after the last round. This value gets collected after each round of simulations to then compare it to other simulations of our model using different start parameters.

To compare different conditions at the start of our simulation, we implemented two different scenarios. In the first one, our initial adopters are selected randomly from all of our agents from different social media platforms. This should represent a condition in which the English word is used by normal users who do not have a position of high network connectivity. In this scenario, we also test how the number of initial adopters influences the speed of the spread and overall adoption of the word. The second scenario has one initial adopter, whom we call an influencer. The influencer is active on all three social media platforms and has a higher connection probability of 0.1 instead of 0.01 for regular agents. This represents a user who has a high connection rate and can share the English word on multiple platforms at the same time. In the end, we want to compare the two conditions to evaluate if we can answer our research question with our model.

3 Results

To evaluate the spread and adoption of English words in our simulated social media network, we ran our model for 100 rounds with 1000 agents and analyzed adoption rates over time across different scenarios. First, we examined how the number of initial adopters affects the overall spread and adoption rate of our model. After that, we compared the platform-specific adoption rates across two scenarios: a random initial adopter and a cross-platform influencer.

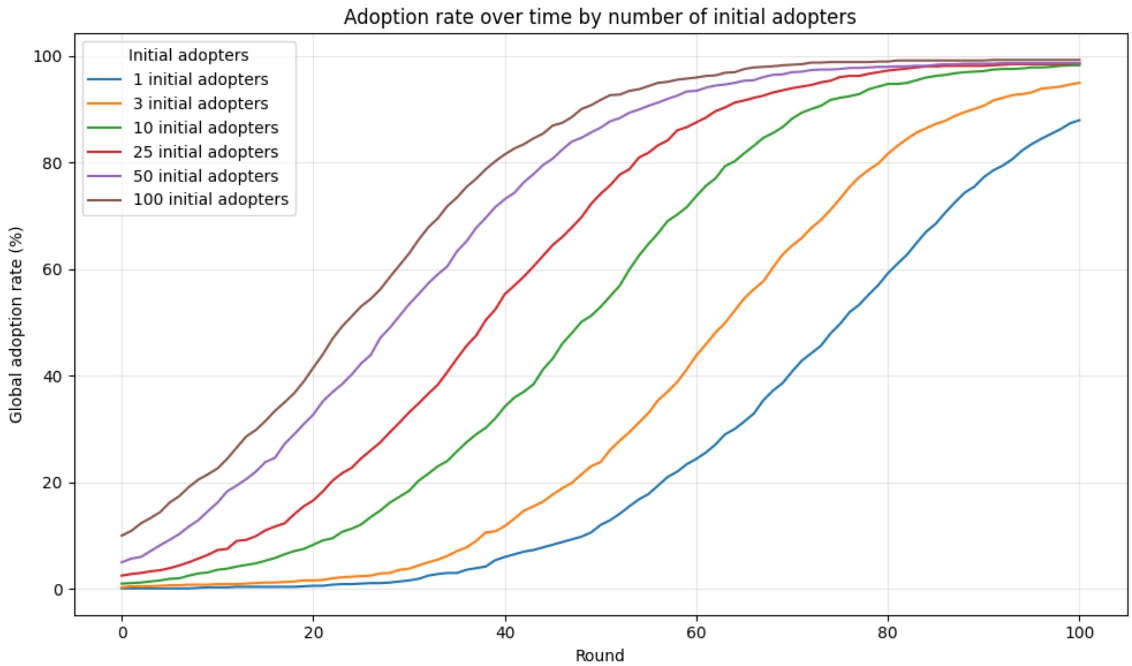

Figure 1: Global adoption rate by number of initial adopters

In Figure 1, we compare the global adoption rate for different numbers of initial adopters, starting at one and going up all the way to 100. We can see a clear relationship between the number of initial adopters and the speed of diffusion. Only one initial adopter gives us a slow adoption rate at the early rounds and a high adoption rate much later than the conditions with many initial adopters. If we look at 50 or 100 initial adopters, we get a rapid early growth and reach a high adoption rate much faster. The curves for 10 or more initial adopters look almost identical, just shifted slightly to the right from 100 initial adopters, so fewer rounds are needed to reach a high adoption rate if we start with more initial adopters. However, given enough rounds, all the conditions reach a high adoption rate, meaning that the final adoption rate does not change much, only the diffusion is faster with more initial adopters.

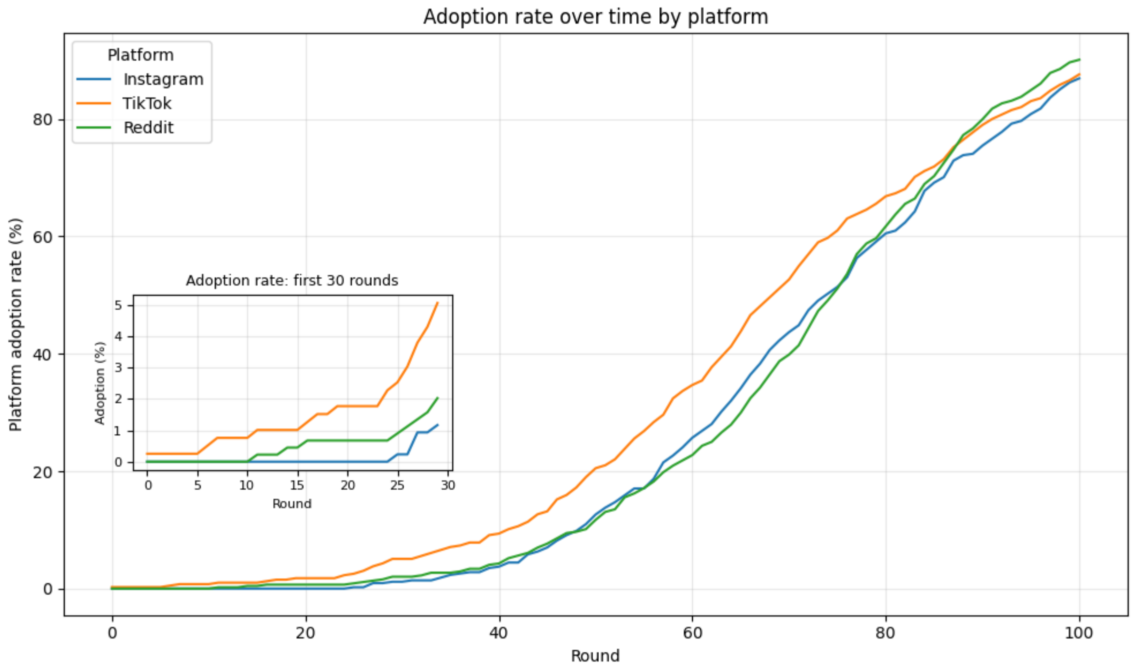

Figure 2: Platform adoption rate with one random initial adopter

Figure 2 shows the adoption rate of every platform individually, with a single random initial adopter. In this case, adoption at the beginning is slow across all three platforms, with only TikTok having any adopters in the first 10 rounds, and the adoption percentage staying under 5% in the first 30 rounds for each platform, as shown in the inset plot. After around round 50, the spreading process accelerates, with each platform showing almost the same rate of change in adoption, while TikTok had a faster rate earlier on because the initial adopter was a TikTok user. By the end of the 100 simulated rounds, each of the three platforms has a very high adoption rate of 85%-90%. This suggests that even though the word started with a single random initial adopter from one single platform, the cross-platform users were able to eventually spread the English word across the entire simulated social-media network.

In Figure 3 we can see the platform adoption rates when we use a cross-platform influencer as our initial adopter. This influencer is active on all three platforms and also has a higher connection probability than

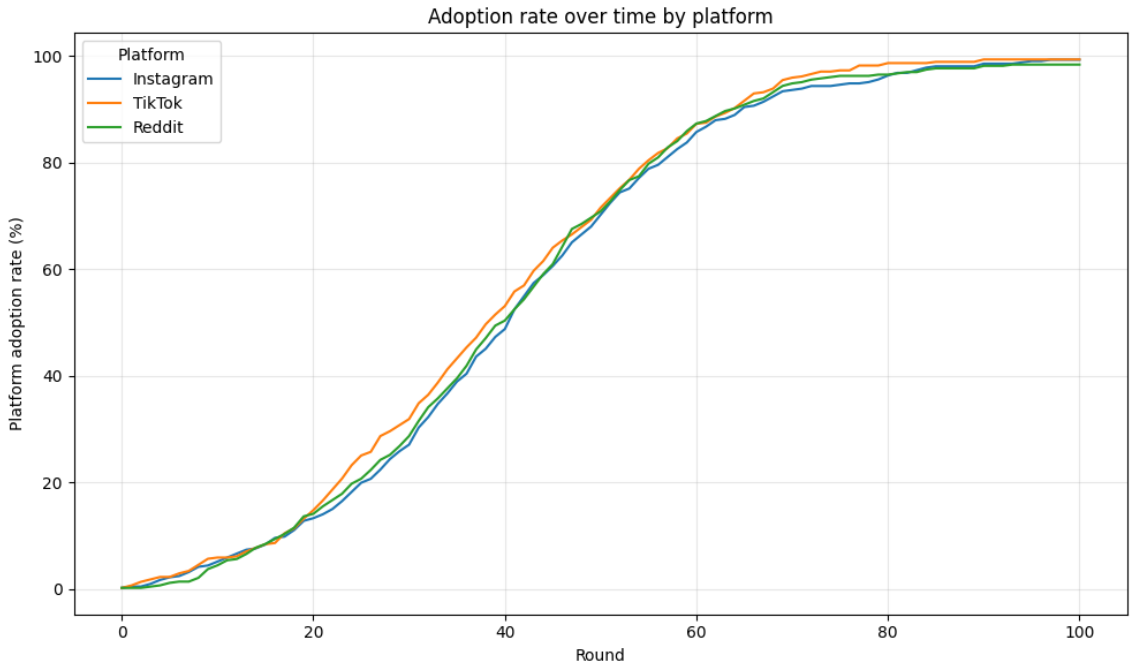

Figure 3: Platform adoption rate with one cross-platform influencer

a normal agent. If we compare this scenario with the one from figure 2, we can see that the adoption spreads much more quickly, and we reach an adoption rate of almost 100% at the end of our simulation across all platforms. The curves of the individual platforms are also very similar throughout the simulation, which indicates that the influencer allows the English word to spread evenly through all the different communities at the same time. This tells us that an influencer can reduce the delay in the adoption rate of different platforms and accelerate the diffusion at the same time compared to a random initial adopter.

Overall, we can say that our initial assumption that cross-platform usage accelerates the spread of English words is supported within our simulation. Figure 1 shows us that having multiple initial adopters in different places in the network gives us a higher adoption rate faster. Figures 2 and 3 further show that the positions of the initial adopters matter, since the cross-platform influencer causes a faster and more synchronized adoption across all platforms than a random initial adopter. Altogether, these results suggest that both the number of initial adopters and a node's position in the network influence how quickly an English word spreads through a simulated German social media network.

4 Discussion

Does cross-platform social media usage accelerate the spread of English loanwords in German social media networks? While running the model with a stronger initial influencer who acts across multiple social media platforms, we can see a faster, more accelerated rate of adoption compared to starting the simulation on distinct platforms. The simulation also shows that having multiple initial adopters across several platforms aids the speed of adoption compared to having a small number of early adopters operating on select platforms.

Across all simulation conditions, the adoption of new English loanwords in our model followed the recognized S-shaped diffusion curve, which is consistent with established empirical studies of language evolution (Nevalainen 2015) and also replicated in Baronchelli et al.'s (2006) agent-based model of language convergence. Changing the number of initial adopters did not change the overall trajectory of adoption, but the more initial adopters we had, the faster the word got adopted. Similarly, if we look at the various platforms, adoption across all social media platforms followed the same similar s-curve shape however we saw slight deviations in how fast the words spread.

Würschinger (2021) discussed that his findings indicated a difference in spread depending on how the neologism was initiated. The community structure during the early stages of diffusion had an impact on the overall success of neologisms. He argues that the spread of new words is generally more successful if their use is not limited to a centralized network of speakers in their early stages. We tested this assumption by comparing the spread of English loanwords first in a centralized manner, assigning a single random initial adopter for each platform individually, and then by assigning a cross-platform influencer as an initial adopter. By comparing both approaches, we were able to demonstrate that the second approach spreads much faster, and the adoption rate across all platforms led to nearly 100% at the end of the simulation, compared to roughly 85-90% for the more centralized approach.

While this study solely focuses on an agent-based model, which is theoretically driven rather than empirically driven, meaningful future work could combine real-world social media data with agent-based modelling for a more representative reproduction of how English loanwords are integrated into the German language. Such an approach could bridge the gap between computational modelling and empirical sociolinguistics. Additionally, researchers in the field of linguistics make a distinction between catachrestic loanwords, which are essentially necessary borrowings such as ‘software,’ and non-catachrestic loanwords, which are adopted although a German equivalent already exists, such as ‘sorry’ (Onysko and Winter-Froemel 2011). Since our model treats all words equally, future work could study this distinction as an additional model parameter to see whether they follow different diffusion trajectories.

When researching anglicisms in the German language, a recurring theme is the ongoing debate surrounding the perceived benefits and downsides of adopting English loanwords into German. Attitudes regarding this matter can range from enthusiastic adoption to active resistance (Hunt 2019). Our model does not account for this since agents adopt words based on exposure and thresholds. Future work could introduce attitude as an additional parameter to investigate how resistance to such phenomena could influence the spread of English borrowings across social media platforms. Investigating language evolution through a digital lens can reveal interesting findings in this day and age, where so much of our communication happens online with rather weak ties as compared to the traditional offline communication in close circles of friends and families. Exploring this with the help of agent-based modelling can give us new tools to simulate these interactions and use the resulting outcomes as a basis for further compelling research in the field of language development.

5 Limitations

For simplicity reasons, our parameters are generally assigned at random, which might not actually be the true representation of German demographics. For example, younger boomers might be over-represented on certain platforms, while higher English proficiency could be under-represented among Gen Z. An additional limitation we face is that all adopted words are treated equally in our model, although there might be a significant distinction in how luxury loanwords spread across social media networks compared to necessary borrowings.

Furthermore, in our model, the adoption of a new English word is binary. We look at whether a word was adopted or not, however, following Coats's (2019) analysis of new English borrowings, this is typically more of a gradual process. New words go through different adaptation stages of morphological and phonological adaptation. He also points out that the majority of anglicisms are rarely utilized, and it's questionable whether they will integrate permanently into the German language.

Lastly, a significant limitation of our model is that we are only looking at how the adoption of new words spreads through online platforms. As Coats (2019) points out in this paper, looking at social media alone may not be representative of how the German language evolves, as much of this happens in other media as well as offline in conversations with friends and family.

6 Conclusion

As the adoption of new English loanwords into the German lexicon Duden continues to grow, exploring the social mechanisms that drives this process becomes increasingly important. This study used an agent-based model to explore how the use of different social media platforms influences the adoption and spread of English words across German social media networks.

The implementation of our model demonstrated that using various social media platforms for our initial adopters led to a faster adoption rate early on, accelerating the spread of new English loanwords across German-speaking social media networks.

In a world where a significant portion of social interactions takes place online, it's important to try to understand how our culture and, for this matter, our language, is shaped by these platforms, which facilitate online communication. Computational modelling allows us to capture the dynamic and social nature of anglicism spread which previous corpus-based approaches struggled. By investigating this, we can contribute to the growing body of research aiming to understand the modern evolution of the German language.

7 References

Baronchelli, Andrea et al. (2006). “Sharp Transition towards Shared Vocabularies in Multi-Agent Systems”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2006.06, P06014. DOI: 10.1088/1742-5468/2006/06/P06014.

Carstensen, Broder (1965). *Englische Einflüsse auf die deutsche Sprache nach 1945*. Beihefte zum Jahrbuch für Amerikastudien 13. Heidelberg: Carl Winter Universitätsverlag, p. 296.

Coats, Steven (2019). “Lexicon geupdated: New German Anglicisms in a Social Media Corpus”. In: *European Journal of Applied Linguistics* 7.1, pp. 31–55. DOI: 10.1515/eujal-2019-0001.

Duden (2024). *Duden: Die deutsche Rechtschreibung*. 29th ed. Berlin: Dudenverlag.

Granovetter, Mark S. (1973). “The Strength of Weak Ties”. In: *American Journal of Sociology* 78.6, pp. 1360–1380. DOI: 10.1086/225469.

Hunt, Jaime W. (2019). “Anglicisms in German: Tsunami or trickle?” In: *Informalization and Hybridization of Speech Practices: Polylingual Meaning-Making across Domains, Genres, and Media*. Ed. by Amei Koll-Stobbe. Peter Lang AG, pp. 25–58. DOI: 10.3726/978-3-653-05414-9. URL: https://hdl.handle.net/1959.13/1417206.

Nevalainen, Terttu (2015). “Descriptive adequacy of the S-curve model in diachronic studies of language change”. In: *Can We Predict Linguistic Change?* Ed. by Christina Sánchez-Stockhammer. Vol. 16. Studies in Variation, Contacts and Change in English. Helsinki: VARIENG. URL: https://urn.fi/URN:NBN:fi:varieng:series-16-4.

Onysko, Alexander and Esme Winter-Froemel (2011). “Necessary loans – luxury loans? Exploring the pragmatic dimension of borrowing”. In: *Journal of Pragmatics* 43.6, pp. 1550–1567.

Poplack, Shana and David Sankoff (1984). “Borrowing: The Synchrony of Integration”. In: *Linguistics* 22.1, pp. 99–135. DOI: 10.1515/ling.1984.22.1.99.

Würschinger, Quirin (2021). “Social Networks of Lexical Innovation: Investigating the Social Dynamics of Diffusion of Neologisms on Twitter”. In: *Frontiers in Artificial Intelligence* 4, p. 648583. DOI: 10.3389/frai.2021.648583.